

Programming Fundamentals Assignment

PW EARNERS : Coding and Web Development

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Variables, Operators, if-else & switch-case

Topics Covered

- Variables
- Data Types
- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Logical Operators
- Increment/Decrement Operators
- if
- if-else
- else-if
- Nested if-else
- switch-case

Not Allowed

The following concepts should **not** be used in this assignment:

- Arrays
 - Objects
 - Functions
 - Loops
 - Classes
 - Advanced concepts
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Section 1 - Variables & Operators

1. Basic Calculator

Create two variables containing two numbers.

Calculate and display:

- Addition
 - Subtraction
 - Multiplication
 - Division
 - Remainder
-

2. Swap Two Numbers

Create two variables:

```
a = 10  
b = 20
```

Swap their values using a third variable.

Expected output:

Before swapping:

```
a = 10  
b = 20
```

After swapping:

```
a = 20  
b = 10
```

3. Student Marks

Create variables for marks of three subjects.

Calculate:

- Total marks
- Average marks
- Percentage

Assume every subject is out of 100.

4. Shopping Bill

Create variables for:

- Product price
- Quantity

Calculate the total bill.

Then apply a discount of **10%** and display:

- Original bill
 - Discount amount
 - Final bill
-

5. Area Calculator

Create variables for length and width.

Calculate:

- Area of rectangle
- Perimeter of rectangle

Use appropriate arithmetic operators.

Section 2 - Comparison & Logical Operators

6. Even or Odd

Take a number and check whether it is even or odd.

Use the % operator.

7. Positive, Negative or Zero

Take a number and determine whether it is:

- Positive
 - Negative
 - Zero
-

8. Greater Number

Take two numbers.

Display:

- First number is greater
 - Second number is greater
 - Both numbers are equal
-

9. Largest of Three Numbers

Take three numbers and find the largest number using `if-else`.

Do not use arrays or any built-in maximum function.

10. Voting Eligibility

Take a person's age.

Rules:

Age $\geq 18 \rightarrow$ Eligible to vote

Age $< 18 \rightarrow$ Not eligible to vote

11. Driving Eligibility

Take:

age

hasLicense

A person can drive only when:

- Age is 18 or above
- They have a valid license

Use a logical operator.

12. Number Range

Take a number.

Check whether the number is between **10 and 100**, including both boundaries.

Example:

Input: 50

Output: Number is between 10 and 100

Section 3 - if, else-if & Nested Conditions

13. Student Grade

Take a student's percentage.

Assign a grade:

90-100 $\rightarrow$ A

80-89 $\rightarrow$ B

70-79 → C
60-69 → D
40-59 → E
Below 40 → F

Also check for invalid percentages below 0 or above 100.

14. Student Result

Take marks of three subjects.

A student passes only when they score **40 or more in every subject**.

If the student passes, calculate the average and display:

Average ≥ 75 → Distinction
Average ≥ 60 → First Division
Average ≥ 50 → Second Division
Otherwise → Pass

If any subject is below 40:

Result: Fail

15. Electricity Bill

Take the number of units consumed.

Calculate the bill using:

0-100 units → ₹5 per unit
101-200 units → ₹7 per unit
Above 200 → ₹10 per unit

Example:

Units = 250

First 100 → 100×5
Next 100 → 100×7
Remaining → 50×10

Display the final bill.

16. Simple Login

Create:

username
password

Correct credentials:

username = "admin"
password = "12345"

If both are correct:

Login successful

Otherwise:

Invalid username or password

Use the logical && operator.

17. Salary Bonus

Take an employee's:

- Salary
- Years of experience

Bonus rules:

Experience ≥ 10 years $\rightarrow$ 20% bonus
Experience ≥ 5 years $\rightarrow$ 10% bonus
Experience ≥ 2 years $\rightarrow$ 5% bonus
Below 2 years $\rightarrow$ No bonus

Calculate and display:

- Original salary
 - Bonus
 - Final salary
-

18. Age Category

Take a person's age.

Display:

0-12 $\rightarrow$ Child
13-19 $\rightarrow$ Teenager
20-59 $\rightarrow$ Adult
60+ $\rightarrow$ Senior Citizen

Also handle invalid negative ages.

Section 4 - switch-case

19. Day of the Week

Take a number from 1 to 7.

Using switch-case:

1 → Monday
2 → Tuesday
3 → Wednesday
4 → Thursday
5 → Friday
6 → Saturday
7 → Sunday

For any other number:

Invalid day

20. Simple Calculator

Take:

- First number
- Second number
- Operator

Supported operators:

+
-
*
/
%

Use switch-case.

Example:

First number: 20
Second number: 5
Operator: *

Output: 100

Also handle division by zero.

21. Month Name

Take a number from 1 to 12.

Use switch-case to display the corresponding month.

Example:

```
1 → January
2 → February
3 → March
...
12 → December
```

For an invalid number:

Invalid month

22. Menu-Based Calculator

Create this menu:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus

Take the user's choice and two numbers.

Use switch-case to perform the selected operation.

23. Traffic Signal

Take a traffic signal color:

```
"red"
"yellow"
"green"
```

Use switch-case.

Display:

red → Stop
yellow → Wait
green → Go

For any other value:

Invalid signal

Section 5 - Mixed Problems

24. ATM Transaction

Create variables:

balance
withdrawAmount

Check:

1. Withdrawal amount must be greater than 0.
2. Withdrawal amount must not be greater than the balance.
3. If valid, subtract the withdrawal amount.
4. Display the remaining balance.

Example:

Balance: ₹10000

Withdraw: ₹3000

Withdrawal successful

Remaining balance: ₹7000

25. Movie Ticket System

Take:

age
numberOfTickets

Ticket prices:

Age below 12 → ₹100

Age 12-59 → ₹200

Age 60+ → ₹120

Calculate the total ticket price.

Example:

Age: 25

Tickets: 3

Total: ₹600

26. Restaurant Menu

Create the following menu:

1. Burger → ₹150
2. Pizza → ₹250
3. Pasta → ₹180
4. Sandwich → ₹120

Take the customer's choice and quantity.

Use switch-case to calculate the total price.

Example:

Choice: 2

Quantity: 3

Total: ₹750

27. Electricity Bill + Discount

Take the electricity units consumed.

Calculate the bill using:

- | | |
|-----------|------------|
| 0-100 | → ₹5/unit |
| 101-200 | → ₹7/unit |
| Above 200 | → ₹10/unit |

Then apply:

- | | |
|---------------|----------------|
| Bill >= ₹2000 | → 10% discount |
| Otherwise | → No discount |

Display:

Units

Original Bill

Discount

Final Bill

28. Simple ATM Menu

Create the following menu:

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

Use switch-case.

Rules:

- Check Balance → Display current balance
 - Deposit → Add money to balance
 - Withdraw → Check whether sufficient balance exists
 - Exit → Display a goodbye message
 - Invalid choice → Display an error message
-

Challenge Questions

29. Number Classification

Take a number.

Determine all applicable information:

- Whether it is positive, negative or zero
- Whether it is even or odd
- Whether it is greater than 100, less than 100, or equal to 100

Example:

Input: 150

Output:

Positive

Even

Greater than 100

30. Student Result System

Create variables for:

studentName
rollNumber
mathMarks
scienceMarks
englishMarks

Calculate:

- Total marks
- Percentage
- Grade
- Pass/Fail

Rules:

Pass/Fail

The student must score at least **40 in every subject**.

Grade

90-100 → A
80-89 → B
70-79 → C
60-69 → D
40-59 → E
Below 40 → F

Display a result like:

STUDENT RESULT

Name : Rahul
Roll No : 101

Math : 85
Science : 78
English : 92

Total : 255
Percentage : 85%
Grade : B
Result : PASS

Total Questions: 30